The documentation and process conversion measures necessary to comply with this revision shall be completed by.18 April 2006.

INCH-POUND

MIL-PRF-19500/313H
18 January 2006
SUPERSEDING
MIL-PRF-19500/313G
14 DECEMBER 2004

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW POWER,
TYPES 2N2432, 2N2432A, 2N2432UB, 2N2432AUB, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for low power, high speed chopper, NPN silicon transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance are provided for die.

1.2 Physical dimensions. See figure 1 (TO-18), figures 2 and 3 (JANHC/JANKC die), and figure 4 (AUB and UB package).

* 1.3 Maximum ratings, unless otherwise specified, T_C =+ 25°C.

Type	V_{CBO} V dc	V_{CEO} V dc	V_{ECO} V dc	I_C mA dc	T_J and T_{STG} °C
2N2432, 2N2432UB	30	30	15	100	-65 to +200
2N2432A, 2N2432AUB	45	45	18	100	

Type	P_T T_A = +25°C (1)	P_T T_C = +25°C (1)	P_T T_{SP} = +25°C	$R_{\theta JA}$ (2)	$R_{\theta JC}$ (2)	$R_{\theta JSP}$ (2)
	mW	mW	mW	°C/W	°C/W	°C/W
2N2432	360	360	N/A	325	150	N/A
2N2432A	360	360	N/A	325	150	N/A
2N2432UB	N/A	N/A	360	N/A	N/A	95
2N2432AUB	N/A	N/A	360	N/A	N/A	95

(1) For derating, see figures 5 and 6.
(2) For thermal impedance curves, see figures 7, 8, and 9.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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* 1.4 Primary electrical characteristics at T_A = +25°C.

Limits	h_{FE1} V_{CE} = 5 V dc I_C = 10 μA dc	h_{FE2} V_{CE} = 5 V dc I_C = 1 mA dc	$h_{FE(inv)1}$ V_{CE} = 5 V dc I_E = 200 μA dc		$V_{CE(sat)}$ I_C = 10 mA dc I_B = 500 μA dc	$r_{ec(on)}$ I_e = 100 μA ac (rms) I_B = 1 mA dc, I_E = 0, f = 1 kHz	
			2N2432	2N2432A		2N2432 2N2432UB	2N2432A 2N2432AUB
Min	30	80	2	3	V dc	Ohms	Ohms
Max		400			.15	20	15

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

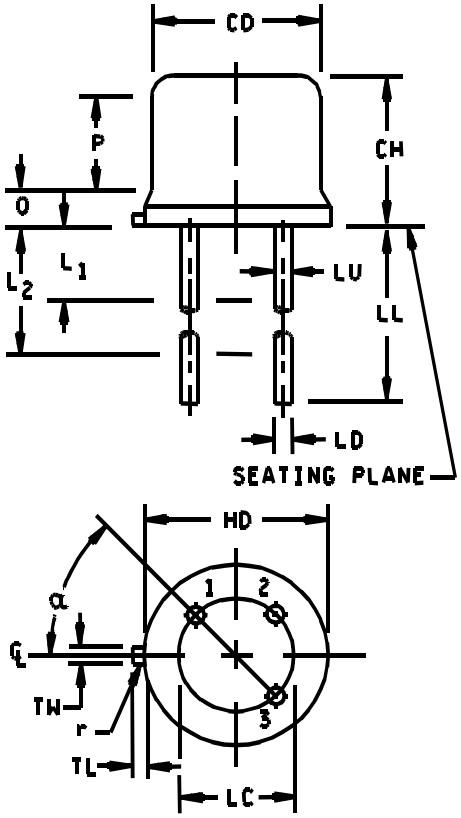
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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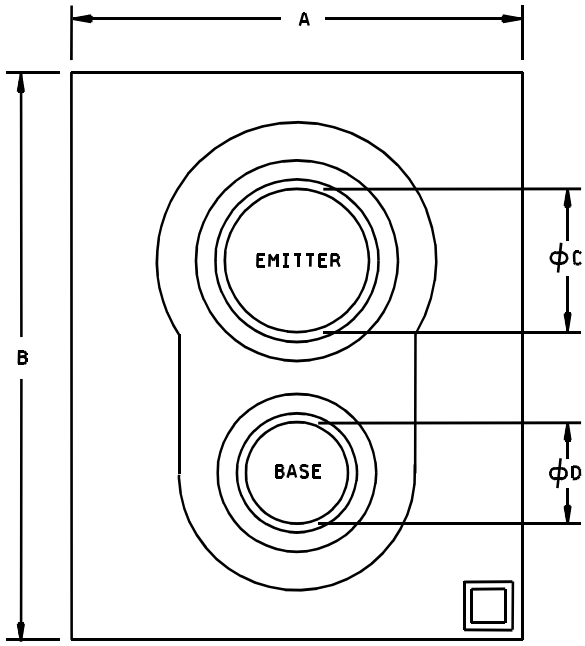
Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8
LU	.016	.019	0.41	0.48	7,8
L1		.050		1.27	7,8
L2	.250		6.35		7,8
P	.100		2.54		
Q		.030		0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r		.010		0.25	10
α	45° TP		45° TP		6



NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (similar to TO-18).

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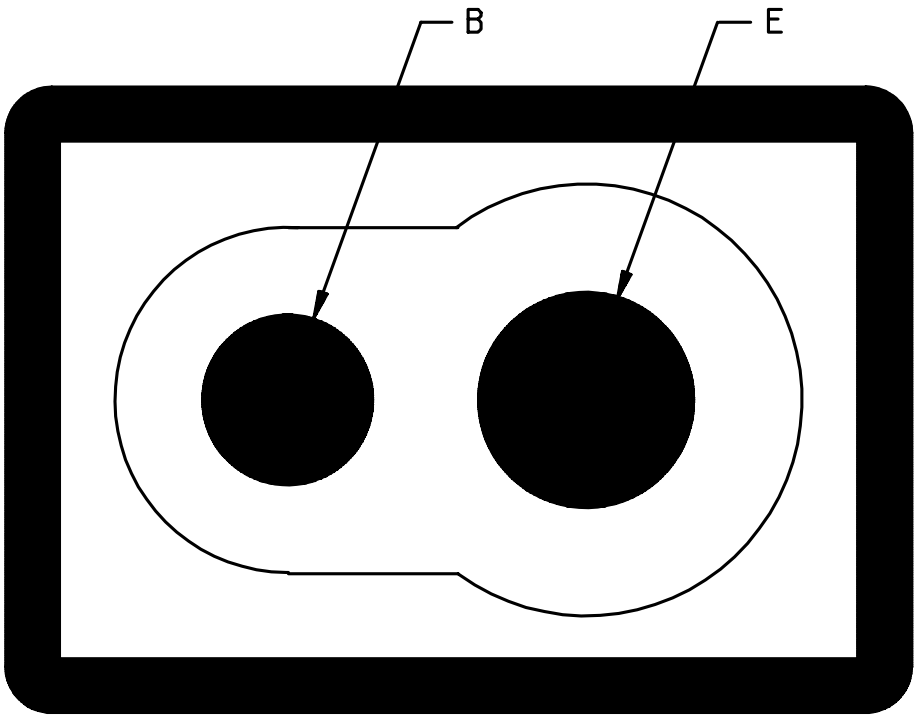


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.013	.017	0.33	0.43
B	.017	.021	0.43	0.53
ϕC		.0048		0.122
ϕD		.0034		0.086

NOTES:
1. Chip thickness: .005 to .007 inch (0.13 to 0.18 millimeter).
2. Top metal: Aluminum 10,000 Å min, 12,000 Å nominal.
3. Back metal: Gold 2,000 Å min, 4,000 Å nominal.
4. Backside: Collector.

FIGURE 2. JANHC (A-version) die dimensions.

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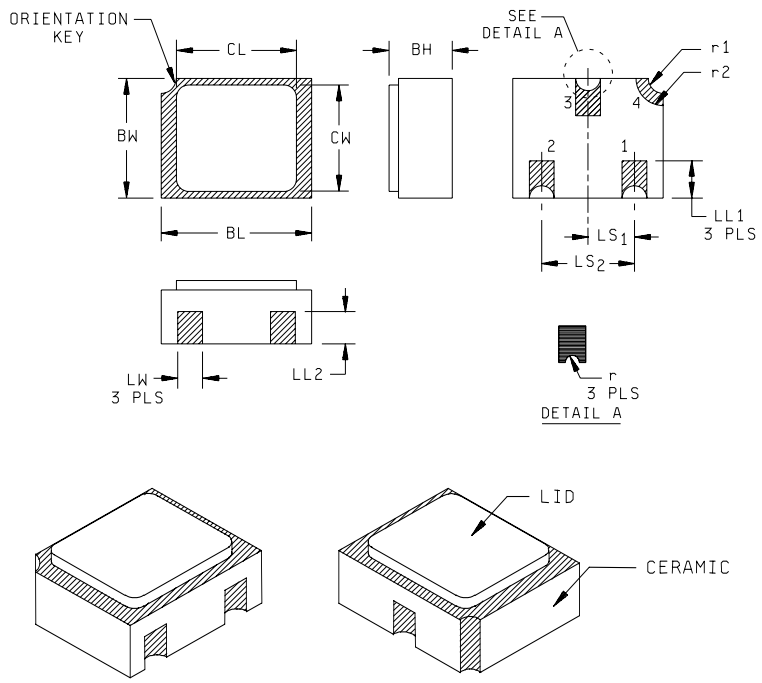


NOTES:

1.	Chip size	.015 x .019 inch ±.001 inch, (0.38 x 0.48 ±0.02 millimeter).
2.	Chip thickness	.010 ±.0015 inch, (0.25 ±0.04 millimeter).
3.	Top metal	Aluminum 15,000Å minimum, 18,000Å nominal.
4.	Back metal	A. Gold 2,500Å minimum, 3,000Å nominal. B. Eutectic Mount - No Gold.
5.	Backside	Collector
6.	Bonding pad	B = .003 inch (0.08 millimeter). E = .004 inch diameter (0.10 millimeter).
7.	Passivation	Si_3N_4 (Silicon Nitride) 2 kÅ min, 2.2 kÅ nom.

FIGURE 3. Physical dimensions, JANHCA and JANKCA (B - version) die.

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
BH	.046	.056	1.17	1.42	
BL	.115	.128	2.92	3.25	
BW	.085	.108	2.16	2.74	
CL		.128		3.25	
CW		.108		2.74	
LL1	.022	.038	0.56	0.96	
LL2	.017	.035	0.43	0.89	
LS1	.036	.040	0.91	1.02	
LS2	.071	.079	1.81	2.01	
LW	.016	.024	0.41	0.61	
r		.008		.203	
r1		.012		.305	
r2		.022		.559	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas
4. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions, surface mount (AUB and UB version).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$h_{FE(inv)}$	Forward current transfer ratio except that the collector and emitter shall be interchanged.
I_e	Emitter current (rms).
PCB	Printed circuit board.
$r_{ec(on}$	Small signal emitter to collector on state resistance.
$R_{\theta JA}$	Thermal resistance junction to ambient.
$R_{\theta JC}$	Thermal resistance junction to case.
$R_{\theta JSP}$	Thermal resistance junction to solder pads.
V_{BC}	Base to collector voltage.
$V_{(BR)ECO}$	Breakdown voltage, emitter to collector, with base open circuited.
$V_{EC(ofs)}$	Emitter to collector offset voltage, i.e., open circuit voltage between emitter and collector when the base to collector junction is forward biased.
V_{ec}	Emitter to collector voltage (rms).

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1, 2, 3, and 4.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.2.1 JANHC and JANKC devices. JANHC and JANKC devices shall be qualified in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance, method 3131 of MIL-STD-750, see 4.3.3.	Thermal impedance, method 3131 of MIL-STD-750, see 4.3.3.
7	Optional	Optional
9	I_{CBO2}, h_{FE1}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO2}; h_{FE1}; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater. Δh_{FE1} = ±15 percent.	I_{CBO2}, h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15 percent.	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = ±15 percent.
14	Required	Required

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc; power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum, T_A ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500 "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

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* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit shall comply with the thermal impedance graph in figures 7, 8, and 9 (less than or equal to the curve value at the same t_H time) or shall be less than the process determined statistical maximum limit as outlined in method 3131 of MIL-STD-750. See table II, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.7 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2 and 4.5.7 herein.

4.4.2.1 Group B inspection (JANS), table VIa of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc.
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table VIa, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve a T_J = +225°C minimum.

4.4.2.2 Group B inspection, (JAN, JANJ, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANJ, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.7.

4.4.3.1 Group C inspection (JANS), table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; (not applicable for UA and UB devices).
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and applied thermal impedance curves.
C6	1026	1,000 hours at V_{CB} = 10 V dc; power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3 n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Group C inspection (JAN, JANJ, JANTX, and JANTXV), table VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UB devices.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3).
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein; delta measurements shall be in accordance with the applicable steps of 4.5.7.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Emitter to collector cutoff current. Method of test shall be in accordance with method 3041 of MIL-STD-750, test condition C, except that all references to the collector and emitter of the transistor under test shall be interchanged.

4.5.3 Emitter to collector breakdown voltage. Method of test shall be in accordance with method 3011 of MIL-STD-750, test condition D, except that all references to the collector and emitter of the transistor under test shall be interchanged.

4.5.4 Forward current transfer ratio (inverted connection). Method of test shall be in accordance with method 3076 of MIL-STD-750, except that all references to the collector and emitter of the transistor under test shall be interchanged. Then $h_{FE(inv)} = I_E / I_B$.

* 4.5.5 Emitter to collector offset voltage. The transistor shall be tested in the circuit on figure 10. The base current shall be adjusted to the specified value. The voltage between the emitter and collector shall be measured using a voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement.

4.5.6 Small signal emitter to collector on-state resistance. The transistor shall be tested in the circuit of figure 11. The base current shall be adjusted to the specified value and an ac sinusoidal signal current, I_e, of the specified rms value shall be applied between the emitter and collector. The rms voltage, V_{ec}, between the emitter and collector shall be measured using an ac voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement. The small signal emitter to collector on-state resistance shall then be determined as follows:

$$r_{ec(on)} = V_{ec} / I_E$$

Where V_{ec} is the rms voltage between the emitter and collector.

* 4.5.7 Delta requirements. Delta requirements shall be as specified below: (1) (2) (3) (4)

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-base cutoff current 2N2432, 2N2432UB 2N2432A, 2N2432AUB	3036	Bias condition D V_{CB} = 25 V dc V_{CB} = 40 V dc	ΔI_{CB02}	100 percent of initial value or 5 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1.0 mA dc; pulsed see 4.5.1	Δh_{FE2}	±25 percent change from initial reading.	

(1) Devices which exceed the table I limits for this test shall not be accepted.
(2) The delta measurements for group B, table VIa (JANS) of MIL-PRF-19500 are as follows: Subgroup 3, 5, and 6; see 4.5.7 herein, steps 1 and 2.
(3) The delta measurements for 4.4.2.2 herein (group B, JAN, JANTX, and JANTXV) are as follows: Steps 1 and 2 of 4.5.7 shall be performed after each step in 4.4.2.2 herein.
(4) The delta measurements for group C, table VII of MIL-PRF-19500 are as follows: Subgroup 6, see 4.5.7 herein, steps 1 and 2 for JANS only.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic seal 4/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = +300°C at t = 2 hrs n = 11 wires, c = 0				
Decap internal visual design verification 4/	2075	n = 4, c = 0.				
Subgroup 2						
Thermal impedance	3131	See 4.3.3	$Z_{\theta JX}$			°C/W
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1).	$V_{(BR)CEO}$			
2N2432, 2N2432UB 2N2432A, 2N2432AUB				30 45		V dc V dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}			
2N2432, 2N2432UB 2N2432A, 2N2432AUB		V_{CB} = 30 V dc V_{CB} = 45 V dc			100 100	µA dc µA dc
Breakdown voltage emitter to collector	3011	Bias condition D; I_E = 100 µA dc; I_B = 0 (see 4.5.3).	$V_{(BR)ECO1}$			
2N2432, 2N2432UB 2N2432A, 2N2432AUB				15 18		V dc V dc
Breakdown voltage emitter to collector	3011	Bias condition D; I_E = 10 mA dc; I_B = 0; pulsed (see 4.5.1 and 4.5.3)	$V_{(BR)EC02}$	10		V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to base cutoff current	3036	Bias condition D	I_{CBO2}			
2N2432, 2N2432UB		V_{CE} = 25 V dc			10	nA dc
2N2432A, 2N2432AUB		V_{CE} = 40 V dc			10	nA dc
Collector to emitter cutoff current	3041	Bias condition C	I_{CES1}			
2N2432, 2N2432UB		V_{CB} = 25 V dc			10	nA dc
2N2432A, 2N2432AUB		V_{CB} = 40 V dc			10	nA dc
Emitter to collector cutoff current	3041	Bias condition C; V_{EC} = 15 V dc; V_{BC} = 0; (see 4.5.2)	I_{ECS1}		2	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 15 V dc	I_{EBO}		2	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 μA dc	h_{FE1}	30		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 mA dc	h_{FE2}	80	400	
Forward-current transfer ratio (inverted connection)	3076	V_{CE} = 5 V dc; I_E = 0.2 mA dc; (see 1.4 and 4.5.4)	$h_{FE(inv)1}$			
2N2432, 2N2432UB				2		
2N2432A, 2N2432AUB				3		
Saturation voltage and resistance	3071	I_C = 10 mA dc; I_B = 0.5 mA dc See 1.4.	$V_{CE(sat)}$		0.15	V dc
Emitter to collector offset voltage		I_B = 200 μA dc; I_E = 0 mA dc (see 4.5.5 and figure 10)	$V_{EC(ofs)1}$			
2N2432, 2N2432UB					0.5	mV dc
2N2432A, 2N2432AUB					0.4	mV dc
Emitter to collector offset voltage		I_B = 1 mA dc; I_E = 0 mA dc (see 4.5.5 and figure 10)	$V_{EC(ofs)2}$			
2N2432, 2N2432UB					1.0	mV dc
2N2432A, 2N2432AUB					0.7	mV dc
Subgroup 3						
High-temperature operation:		T_A = +125°C				
Collector to emitter cutoff current	3041	Bias condition C	I_{CES2}			
2N2432, 2N2432UB		V_{CE} = 25 V dc			250	nA dc
2N2432A, 2N2432AUB		V_{CE} = 40 V dc			250	nA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Emitter to collector cutoff current	3041	Bias condition C; V_{EC} = 15 V dc; V_{BC} = 0 (see 4.5.2)	I_{ECS2}		200	nA dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 µA dc; pulsed (see 4.5.1)	h_{FE3}	10		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 mA dc; pulsed (see 4.5.1)	h_{FE4}	25		
Forward-current transfer ratio (inverted connection)	3076	V_{CE} = 5 V dc; I_E = 200 µA dc; pulsed (see 4.5.1 and 4.5.4)	$h_{FE(inv)2}$	1.8		
Subgroup 4						
Small signal emitter collector on state resistance		I_B = 1 mA dc; I_E = 0; I_e = 100 µA ac (rms); f = 1 kHz (see 4.5.6 and figure 11)	$r_{ec(on)}$			
2N2432, 2N2432UB					20	Ω
2N2432A, 2N2432AUB					15	Ω
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 20 MHz	$\|h_{fe}\|$	2	10	
Open circuit output capacitance	3236	V_{CB} = 0; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		12.0	pF
Input capacitance	3240	V_{EB} = 0 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		12.0	pF
Subgroups 5 and 6						
Not applicable						
Subgroup 7 4/						

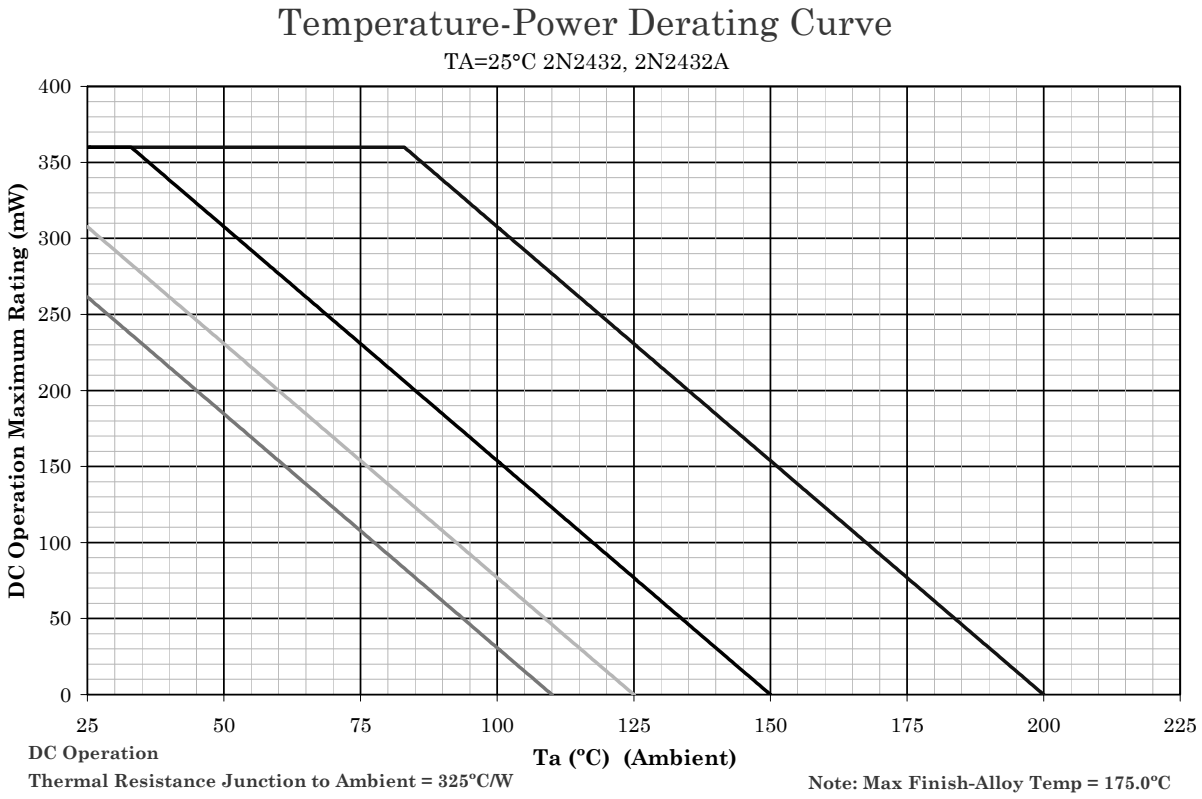
1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not applicable for JANS devices.
5/ Not required for laser marked devices.

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* TABLE II. Group E inspection (all quality levels) - for qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.5.7 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	V_{CB} = 10 V dc, 6,000 cycles, forced air cooling allowed on cooling cycle only.	
Electrical measurements		See table I, subgroup 2 and 4.5.7 herein.	
Subgroup 4			
Thermal resistance	3131	$R_{\theta JSP}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets).	15 devices, c = 0
Thermal impedance curves		Each supplier shall submit a thermal impedance ($Z_{\theta JX}$) histogram of the qual lot and thermal impedance curves of the best device and the qual lot average. A histogram of thermal resistance ($R_{\theta JX}$) is also required (116 devices). In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report for approval. The approved thermal impedance conditions and limit for $Z_{\theta JX}$ shall be used by the supplier in screening and for endpoint measurements as applicable. The approved thermal impedance conditions for $R_{\theta JX}$ shall be used by the supplier for conformance inspection.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 6			3 devices
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B	

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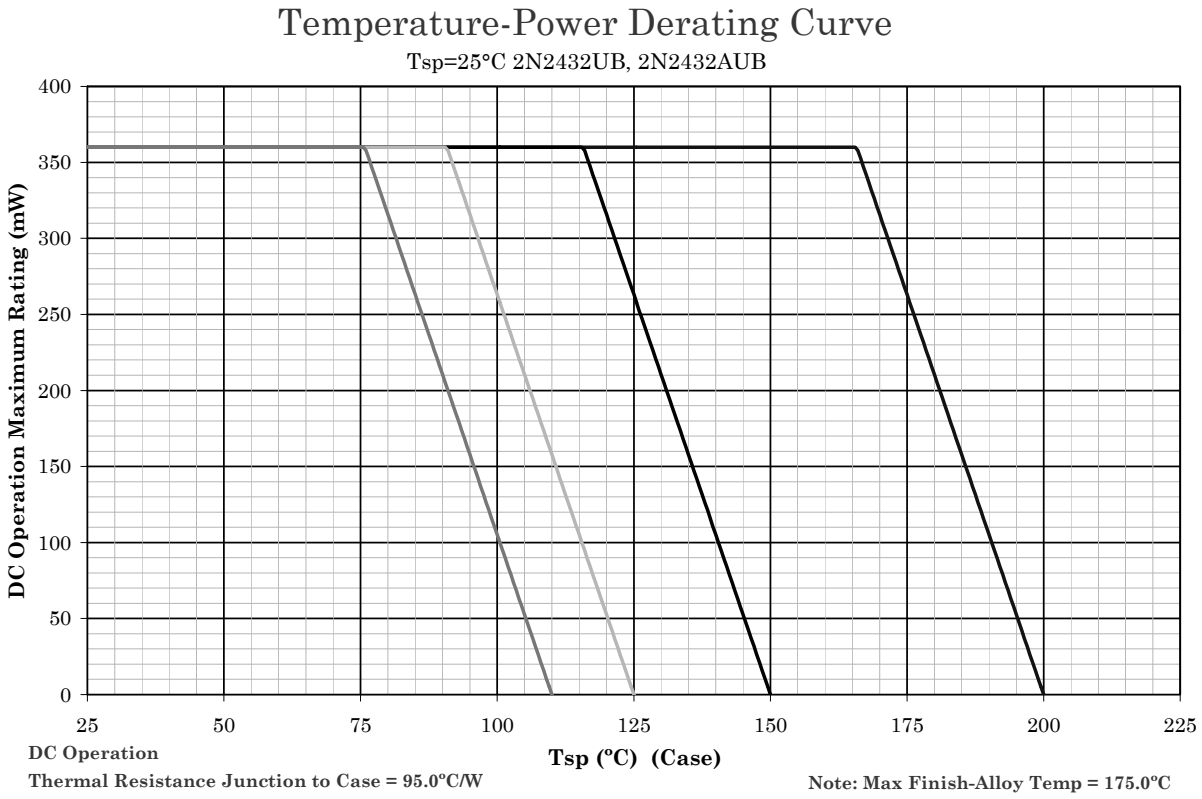


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 5. Derating for all devices ($R_{\theta JA}$) for 2N2432 and 2N2432A.

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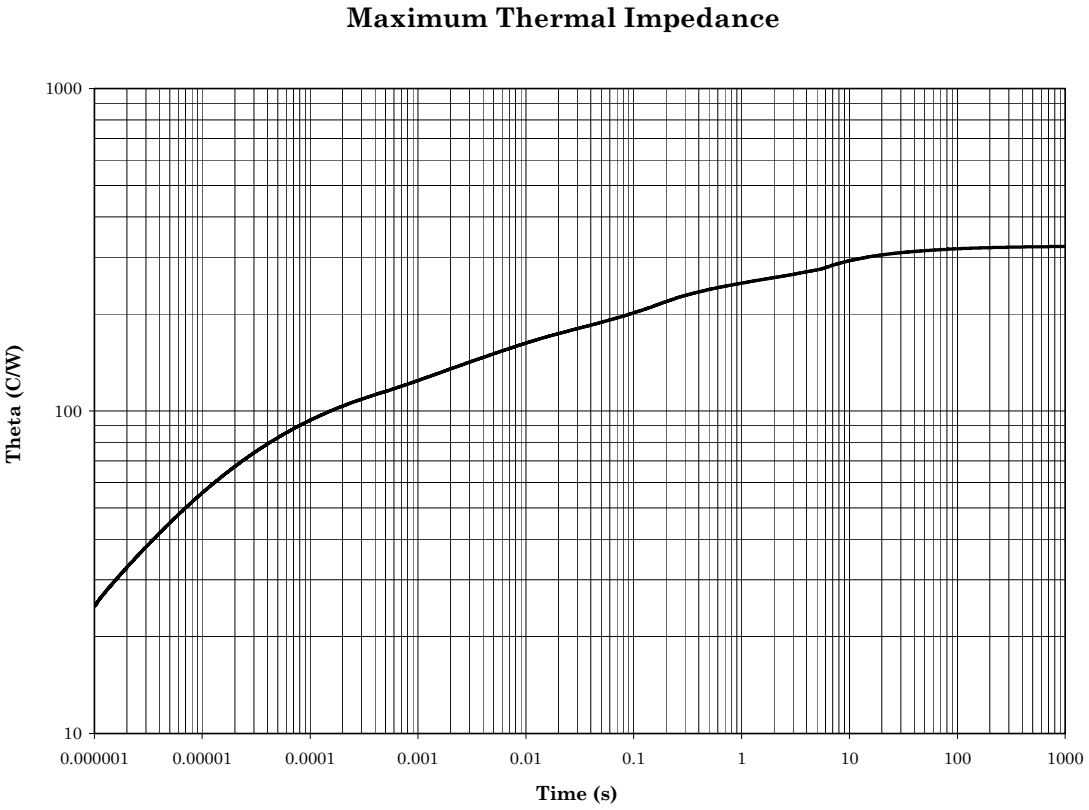


NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit T_J in their application.

* FIGURE 6. Derating for all devices ($R_{\theta JA}$) for 2N2432UB and 2N2432AUB.

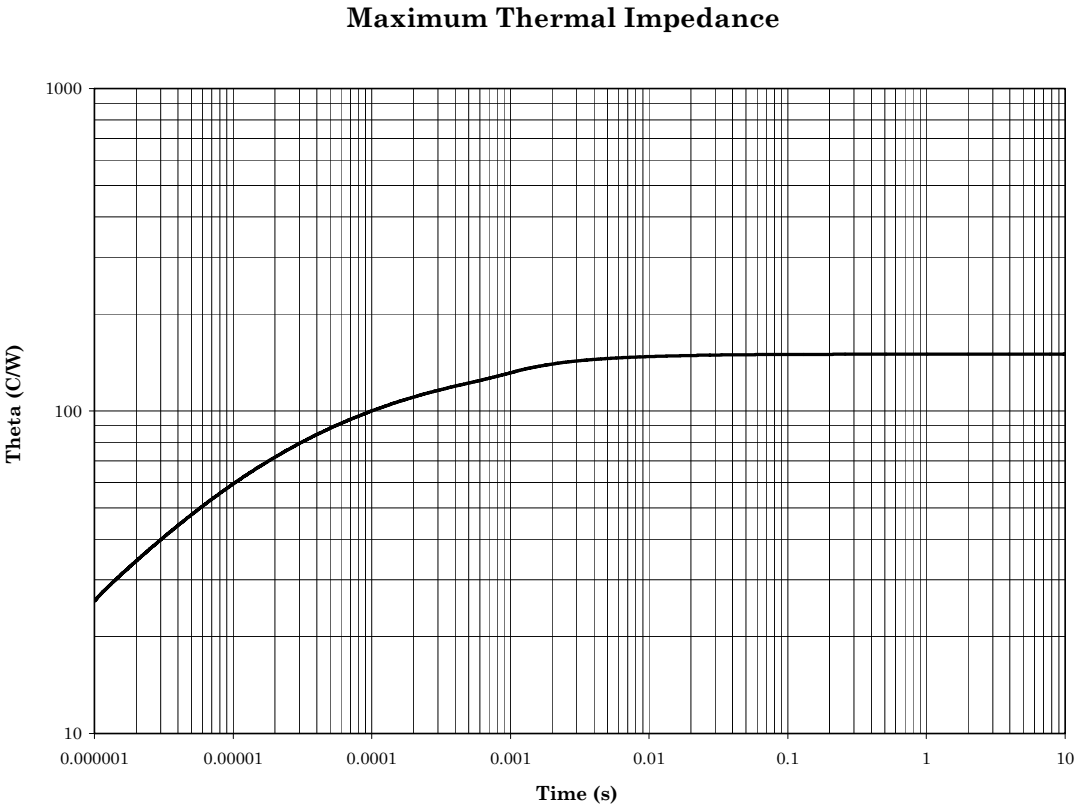
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Resistance $R_{\theta JA}$ = 325°C/W.

FIGURE 7. Thermal impedance graph ($R_{\theta JA}$) for 2N2432 and 2N2432A.

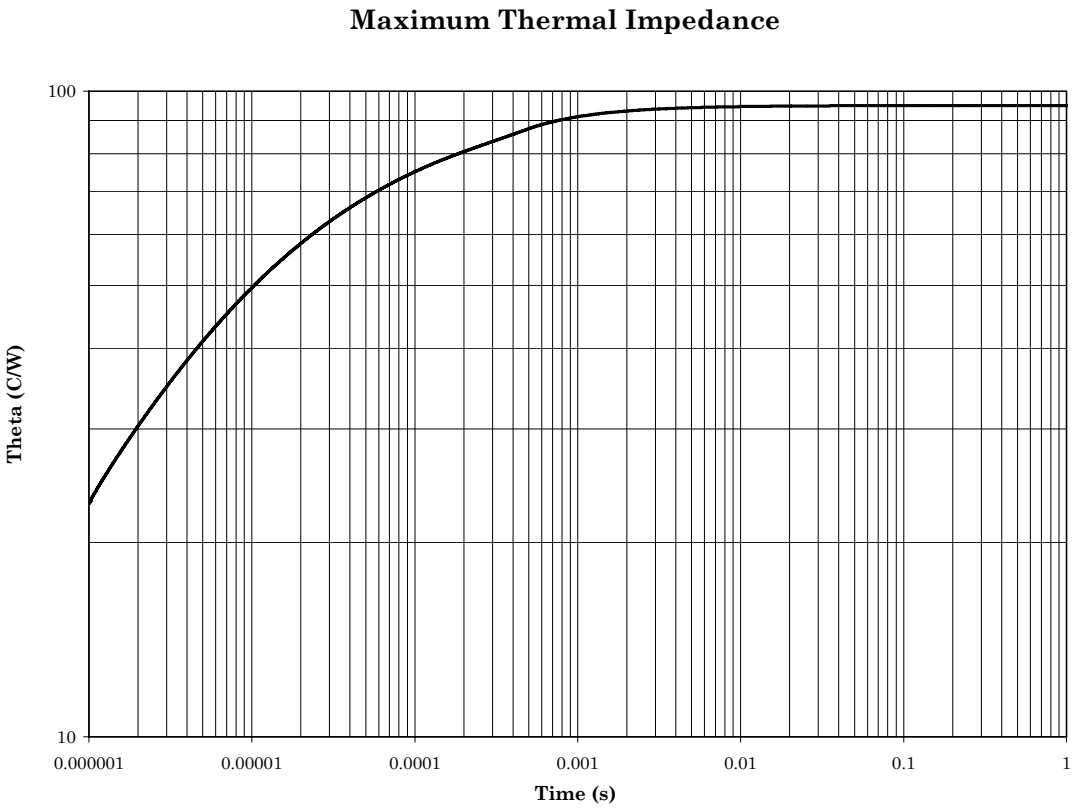
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Resistance $R_{\theta JC}$ = 150°C/W.

FIGURE 8. Thermal impedance graph ($R_{\theta JC}$) for 2N2432 and 2N2432A.

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Resistance $R_{\theta JSP}$ = 95°C/W.

FIGURE 9. Thermal impedance graph ($R_{\theta JSP}$) for 2N2432UB and 2N2432AUB.

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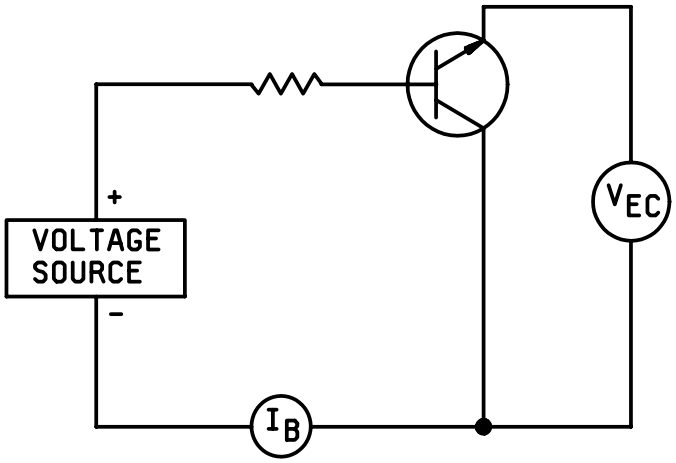


FIGURE 10. Emitter to collector offset voltage test circuit.

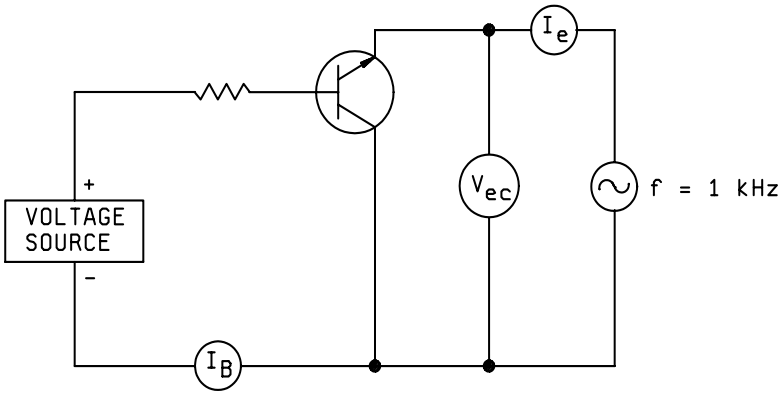


FIGURE 11. Small-signal emitter-collector on-state resistance test circuit.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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